EthicalAI: An Evidence-Linked Catalogue for AI Integrity

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Abstract

EthicalAI organises integrity failures and related behavioural patterns as definitions linked to evidence, obligations, harms and neighbouring concepts. This paper reports the construction and review of its historical 207-record study. The construction log classified 186 records as grounded in worked examples from source papers and retained 21 without that level of support. A citation audit recorded 29 repaired source mismatches among roughly fifty papers examined. A subsequent checking pass reported hard errors in 26 of the 186 grounded records and minor issues in 42. These are findings about the reviewed artifact, not error rates for AI research generally. A deterministic 33-record codebook check produced model-to-catalogue Cohen kappa values of 0.90 and 0.83; this is a model-based consistency check, not human validation. Twelve author-selected public incidents illustrate the vocabulary's use, including one partial fit. The later 434-record release adds 227 untested analogies and a broader bibliography, which are outside those historical evaluation denominators. The contribution is a reusable reference with an explicit account of its construction errors and evidence limits, rather than a validated set of distinct AI mechanisms.

1. Introduction

An AI assistant can cite a nonexistent case, retain a user's mistaken framing after finding contrary evidence, or extend a coding task without disclosing the extra changes. Each failure calls for a different check. Citation resolution does not test whether the assistant updated its framing; correct framing does not establish that an action stayed within the agreed scope. A catalogue can make these distinctions available to a reviewer before they become a checklist, an experimental label or an audit obligation.

EthicalAI collects such distinctions in one reference. Entries include a stable identifier, an operational definition, evidence pointers, examples and typed relations. Human cognitive concepts supply part of its vocabulary. The parallel is a way to formulate a candidate behavioural test, not evidence that an AI system has the corresponding human mental process. Other entries name attacks, system properties or outcomes. These different kinds of record are useful to an auditor, but their inclusion does not make them interchangeable scientific constructs.

This paper examines the historical 207-record catalogue and the checking work documented during its construction. It asks what the resource represents, what its evidence supports, and what went wrong when examples were attached to sources. Two smaller checks examine category-label recoverability and use on public incidents. The companion advisory-auditing paper studies a separate question: whether focused checks of named obligations help review coding-agent drafts. Its six-clause pilot does not validate the catalogue as a whole.

The current release contains 434 records. Section 6 separates that later expansion from the original study so that more entries and citations do not silently enlarge the evidence claimed here.

2. Related resources

Existing resources provide overlapping views of AI failures. The MIT AI Risk Repository synthesises 74 frameworks containing 1,725 risks and offers causal and domain taxonomies [Slattery et al. 2024]. The NIST AI Risk Management Framework describes trustworthiness and governance practices [NIST 2023]. These resources organise risks and responsibilities at a broader level than an individual draft's evidence relationships. Their scope also includes causes and AI-originated failures, so they should not be reduced to lists of downstream harms.

Application-security resources overlap directly with this catalogue. The OWASP Top 10 includes prompt injection, excessive agency and misinformation [OWASP 2025]. Some of these failures involve an adversary and others need not. EthicalAI reuses these concerns within a vocabulary that also includes memory, reasoning, confidence and task-scope patterns. Its partial crosswalk records proposed correspondences; it does not establish equivalence between frameworks.

Behavioural benchmarks provide the closest comparison. Malberg et al. [2024] evaluate 30 cognitive biases across 20 models, with 30,000 tests. EthicalAI's classic-bias subset overlaps that work. Hallucination surveys and benchmarks, including Ji et al. [2022] and HALoGEN [Ravichander et al. 2025], also distinguish several failure types within a family. These are measurement resources as well as classifications. The larger number of EthicalAI records is a coverage choice, not evidence of stronger measurement or more distinct mechanisms.

The resource combines definitions and source records across these areas and connects them to an advisory audit vocabulary. Its proposed value is that a reviewer can find adjacent concepts and inspect the evidence behind an example without assembling several separate lists. The historical checks below assess parts of that reference; they do not establish that its organisation is uniquely correct or that another taxonomy could not support the same uses.

3. Representation and construction

3.1 Records and categories

The historical catalogue contains 207 records. Its organising scheme assigns 168 to ten cognitive domains and 39 to an AI-native category. The domains are labels for navigating the catalogue, not claims about brain localisation or shared computational mechanisms.

Category	Historical records
Perception	11
Memory	13
Attention	12
Heuristics	25
Reasoning	22
Metacognition	16
Language	14
Social	28
Moral	12
Self and agency	15
AI-native	39

The legacy AI-native label groups failures formulated in terms of machine systems, such as prompt injection and reward tampering. It should not be read as proof that no human analogue exists. Reward hacking and scope creep have recognisable human counterparts. The exclusive assignment is an operational choice in the released schema, and its boundaries require further study.

Each record carries a stable slug and reference number, a definition, examples, source pointers and relations. A number identifies a record rather than a distinct mechanism. Synonyms, narrower variants and linked overlaps therefore need explicit representation rather than being counted as independent findings. The historical priority set highlights 26 records for auditing; it is an author-selected worklist rather than a measured ranking of failure frequency.

3.2 Relation semantics

The graph distinguishes several relationships. `violates` points from a mode to a catalogue obligation. `causes` points to a harm type, such as a wrong answer, wasted effort or loss of oversight. Despite that field name, these edges record author-judged harm associations, not estimated causal effects. `relates_to` connects neighbouring modes; subtype and overlap relations express other conceptual distinctions. Partial `maps_to` records connect entries with external frameworks.

The released `causes` relation does not connect one mode to another. Consider the dealership-chatbot incident discussed in Section 5. An analyst can describe it as an instruction attack followed by compliant output and an unauthorised offer. That is an interpretation of the reported sequence, not a three-mode causal path encoded by the present graph. It would be possible to add a distinct relation for event sequences, but its meaning and evidence would need to be specified. The existing graph supports navigation and comparison; it does not demonstrate how one failure produces another.

3.3 Evidence and examples

The construction procedure sought a source passage for each worked example, including its numbers, model names and experimental conditions. The historical log records full-text inspection and inline arXiv tags for those examples. A plausible scenario without a matching reported finding was retained as an illustration rather than treated as a measured result. The log classified 186 records as grounded under this procedure and held 21 without that level of model-side support. A failed search means support was not established in the checked material; it does not show that no relevant paper exists.

Source identity, topical relevance and support for a particular claim are different checks. A psychology article can define a human construct without testing an AI analogue. A theoretical article can introduce a useful label without establishing its mechanism. Even an on-topic AI experiment may not support every example or causal interpretation attached to the record. The evidence relationship should therefore specify both what was consulted and what it supports: a definition, a human finding, an AI finding, an interpretation or contrary evidence.

Package hallucination illustrates the importance of preserving the denominator. Spracklen et al. [2025] generated 576,000 code samples and evaluated the packages recommended through their extraction procedures. They classified 440,445 of about 2.23 million recommended packages as hallucinations, or 19.7%. That figure is a rate over package recommendations, not the percentage of code samples containing a hallucination. It is also specific to the tested models, prompts and package-registry snapshot.

4. Findings from construction

4.1 Citation mismatches

The running citation audit documents cases in which a plausible reference did not support the catalogue's claim. Among roughly fifty papers examined during grounding, 29 mismatches were repaired by changing the cited source. Other mismatches led to records being held without a grounded AI example. The approximate denominator and the distinction between repaired and held records prevent treating 29 as an exact overall error rate.

The recorded problems include papers on a different phenomenon, a result in the opposite direction, and an appropriate test with a null finding. For example, one source attached to model agency concerned a human user's perceived control. A source used for pragmatic over-accommodation instead reported under-use of pragmatic inference. Some proposed bias analogies were tested without the expected effect. These are different reasons to revise a claim, and a bibliography entry alone does not distinguish them.

The audit supports a practical construction requirement: attach a claim to its supporting passage and experimental scope, not just to a title or a search hit. It does not estimate how often other catalogues contain such errors, because the checked papers were not a random sample of research publications.

4.2 Rechecking worked examples

The subsequent verification report describes four checking batches covering the 186 historically grounded records and approximately 510 tagged examples. The record, rather than a new full-text sweep for this revision, is the source of the following counts.

Recorded outcome	Records	Share of the 186-record cohort
No issue detected	118	63%
Minor issue	42	23%
Hard error	26	14%

A hard error meant that a figure was contradicted by, or absent from, the cited paper. Minor issues included loose attribution, rounding and confusion between experimental conditions. The report records no unreachable papers in this cohort. These are outcomes of the checking procedure, not independently adjudicated labels or a residual error estimate for the corrected release.

Several corrections changed a number or its interpretation: a city name, a percentage, a measurement scale, or the observed and hidden reward labels in an example. Six hard-error cases concerned quoted examples that were not found in their cited papers. The record describes replacing these with supported material or marking them as untagged illustrations. It records corrections to all 68 flagged entries, while leaving open whether an independent subsequent reader would find further errors.

The two audits address different questions. Citation checking asks whether the source concerns and supports the claimed phenomenon. Example checking asks whether a particular statement is faithful to that source. Their combined lesson is narrower than a general claim about AI reliability: this artifact needed both checks, and initial source inspection did not prevent later transcription and interpretation errors.

5. Checks of category use and incident mapping

5.1 A model-based codebook check

The codebook study used 33 records, three from each category. The sampling script selects the first three records by reference number in each group, so the sample is deterministic and balanced rather than random. Two separate language-model calls received category descriptions and each record's name and definition, without its stored category. Agreement was measured against the catalogue and between the calls.

The sonnet rater agreed with the catalogue on 30/33 records, with Cohen kappa 0.90. The haiku rater agreed on 28/33, with kappa 0.83. Agreement between the raters was kappa 0.93. There were eight rater-to-catalogue disagreements across five records; three records were assigned differently by both raters. These included cognitive miserliness, mind-wandering and scope creep, each near a boundary between the supplied categories.

This tests whether two model calls can apply the codebook similarly to its author. It does not establish human inter-annotator agreement, construct validity or reliable categorisation outside this small, ordered sample. Shared training and shared assumptions can make model agreement less informative than agreement from independent human annotators.

5.2 An incident-mapping exercise

The historical coverage record maps twelve author-selected, publicly reported incidents from 2023 to 2025 to catalogue entries. The record includes source fabrication in legal filings, an invented airline refund policy, instruction attacks on customer-service bots, and harmful or misleading conversational output. A case resting on allegations was set aside from the counted set.

All twelve received at least one existing label in the author's mapping. Three were legal-citation incidents assigned to source fabrication; that concentration describes this selected sample, not the prevalence of failures in deployment. Several incidents received multiple labels. Assigning those labels can describe different aspects of the same episode without establishing a causal connection between them.

The mapping for AI Overviews' satirical food and health advice was partial. A hallucination label covered the false answer, while the available labels did not cleanly capture treating a joke or satirical source as authoritative. The record proposed source-genre credulity as a possible refinement. This is a candidate distinction arising from the exercise, not evidence that it is a new, independent mechanism.

Because the incidents were selected by the author and mapped manually, the exercise demonstrates possible uses of the vocabulary. It cannot establish coverage of future incidents or superiority to another taxonomy. A stronger test would specify the sampling frame and matching rules in advance and have independent annotators score both fit and failure to fit.

6. Release versions and reuse

The historical study and the current catalogue have different scopes. The 207-record study is retained as a versioned baseline; the current version 1.3 contains those records plus 227 proposed human-behaviour analogies, for 434 in total. None of the additional records enters the denominators of 186 checked records, 33 codebook items or twelve incidents above. Nor do the companion method paper's six-clause pilot results evaluate those 227 analogies.

The November 9, 2026 bibliography snapshot has 1,175 selected citation bindings across 1,036 publication identifiers. Of the 434 records, 359 have three selected references and 75 have fewer, including nine with none. These are bibliography availability counts. They are not counts of independently confirmed mechanisms or replications. Preprints, theoretical articles, empirical studies and reviews must be interpreted according to their individual access and relevance notes.

The current row totals are separately recorded arXiv metadata-query matches in cs.AI, cs.CL and cs.LG. They are not full-text citation counts or an exhaustive count of relevant research. Keyword matches can include unrelated uses, and human research cited in a record can fall outside that AI-domain query. The query, filters and retrieval date travel with the total.

Stable slugs and reference numbers permit reuse across releases. A retained overlap or retired concept should keep its identifier with an explicit relationship, rather than being silently deleted or renumbered. Consumers can use the catalogue for retrieval, annotation proposals or advisory audits while retaining the source passage and evidence status of each claim.

7. Limitations

The construction and checking records are a single project's audit trail. They do not provide an independent human replication of the 186 grounding decisions, the hard-versus-minor classification or the corrected examples. The reported 26/186 hard-error proportion belongs to that historical cohort and procedure. It should not be used as a population estimate for the present 434 records, other catalogues or AI-generated research generally.

The catalogue mixes behaviours, properties, attacks and outcomes, and its categories and links reflect author judgments. Shared labels do not establish shared mechanisms between people and AI. Clinical terms do not diagnose a model's feelings, unconscious motives or intent. Likewise, a source's score for internal reasoning and a model's output accuracy are different measurements; their divergence alone does not demonstrate deliberate deception.

The model-based codebook check cannot substitute for blinded human annotation. The incident exercise uses salient public cases and a permissive matching task. Neither measures how accurately a reviewer can distinguish close alternatives, how often the catalogue misses an externally sampled failure, or whether its graph improves auditing over a flat list. Those questions require separate evaluation. The resource makes no completeness or state-of-the-art detection claim.

8. Availability and reproducibility

The machine-readable catalogue is `ontology/ethical-ai.yaml`, with field and relation semantics in `ontology/SCHEMA.md`. The historical version-1.0 catalogue is pinned at commit 8798134; the later version-1.3 bibliography described in Section 6 is pinned at commit 3b9b500. These snapshots have different evidence cohorts and should not be interchanged when reproducing a count.

The historical evidence is in `experiments/citation-audit.md`, `experiments/verification-report.md`, `experiments/agree-gold.json`, the taxonomy-agreement outputs, and `experiments/coverage-incidents.md`. The eligibility rule in `experiments/verify-sample.py` selects records with at least three examples; that mechanical selection is not an independent test that the examples are grounded. The later source-review history is retained under `references/`, including the November 9 literature-count and citation review.

`CITATION.cff` records the citation metadata. Licensing and archival-release steps are described in `RELEASE.md`; a live website or a changing branch should not replace the immutable artifact cited for a reported result.

9. Conclusion

EthicalAI provides definitions, evidence pointers and conceptual relations for reviewing AI integrity. Its historical construction records show why a plausible citation and an accurate worked example need separate checks. The 207-record study also supplies small demonstrations of codebook use and incident mapping, with their sampling and annotation limits stated. The later bibliography and 227 untested analogies extend the reference without extending those empirical claims. The resource is useful to the extent that readers can inspect and correct the relationship between a claim, its source and the scope in which it was observed.

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Appendix A. Example records

The following records illustrate the distinction between an entry, a reported finding and a proposed interpretation. The source-fabrication scenario is an untagged illustration: an assistant cites a plausible legal case that does not exist. It describes the operational failure without reporting an experiment.

The package-hallucination example in Section 3 is a source-specific measured result. Its numerator, denominator, tested model set and registry assumptions must remain attached to the claim. The failure definition can apply more widely than that experiment, but the experiment's percentage cannot.

Reward hacking is retained in the legacy AI-native category. That placement does not deny human analogues such as gaming a performance target. Its harm links express catalogue judgments and its relationship to other modes is not a causal result about the sequence of a particular incident.